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(54) **SYSTEMS AND METHODS FOR DELIVERY OF CHORDAE REPLACEMENT SYSTEM**

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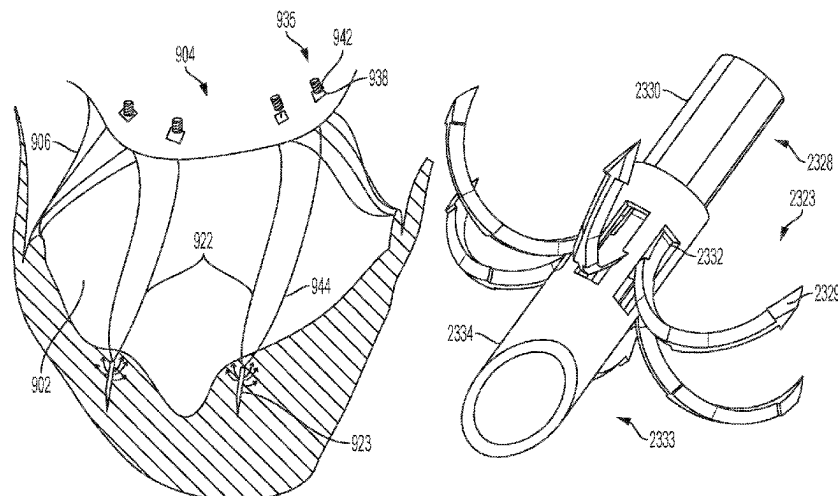
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(57)

ABSTRACT

A system and method of repairing a native chordae of a patient using an artificial chordae. The artificial chordae is inserted into the patient in a non-deployed configuration using a delivery system and is delivered to a desired position within the patient. An operator causes the artificial chordae to transition from the non-deployed configuration to a deployed configuration. The artificial chordae is anchored to a myocardium of the patient. The artificial chordae is attached to a leaflet of the native chordae at an attachment location and tuned to a desired tension.

13 Claims, 23 Drawing Sheets



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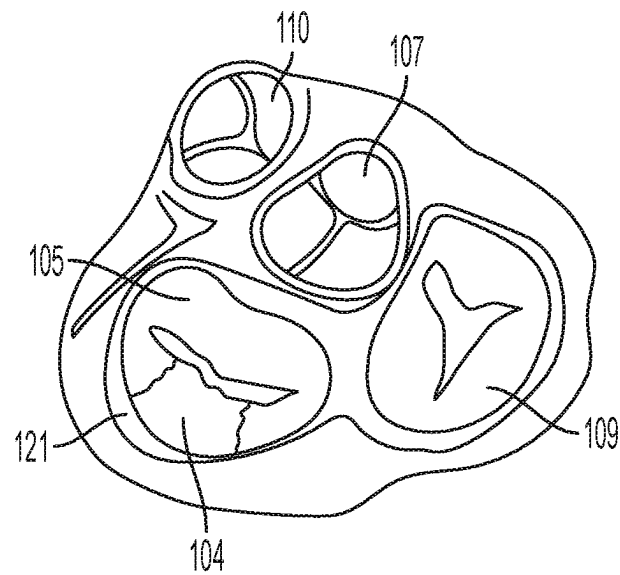


FIG. 1

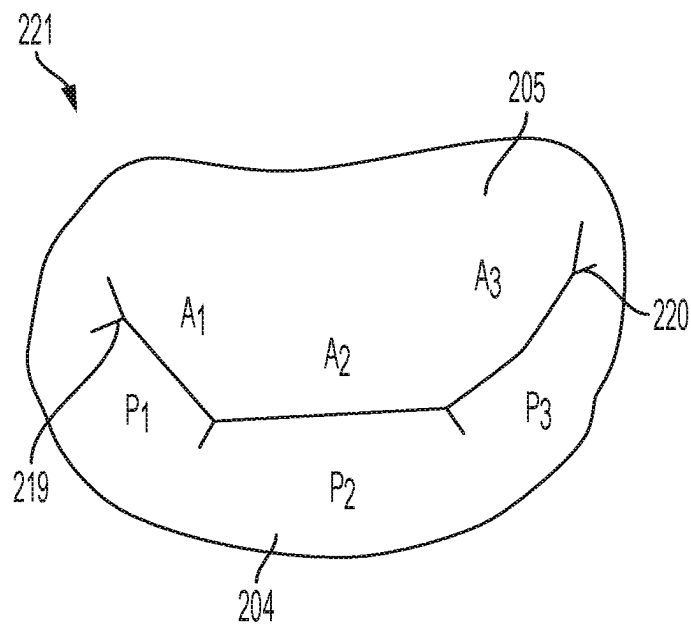


FIG. 2

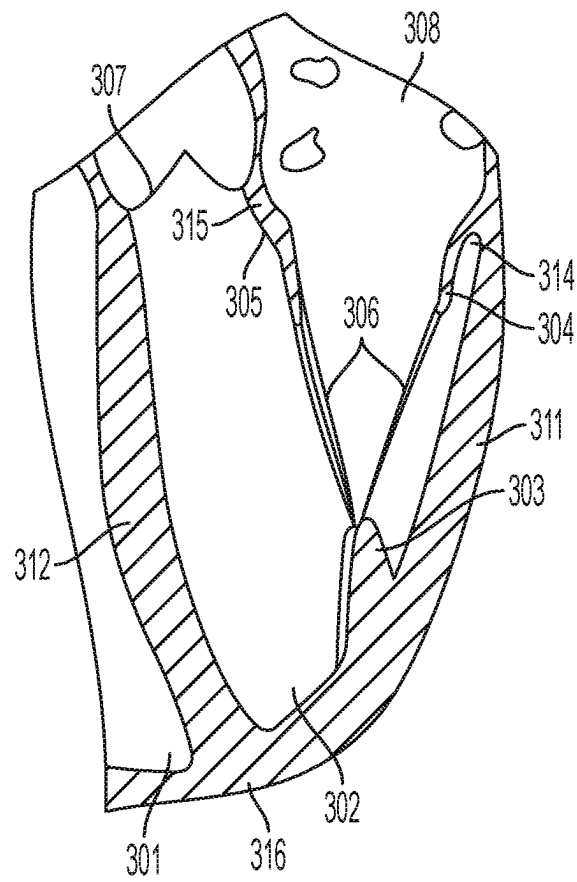


FIG. 3

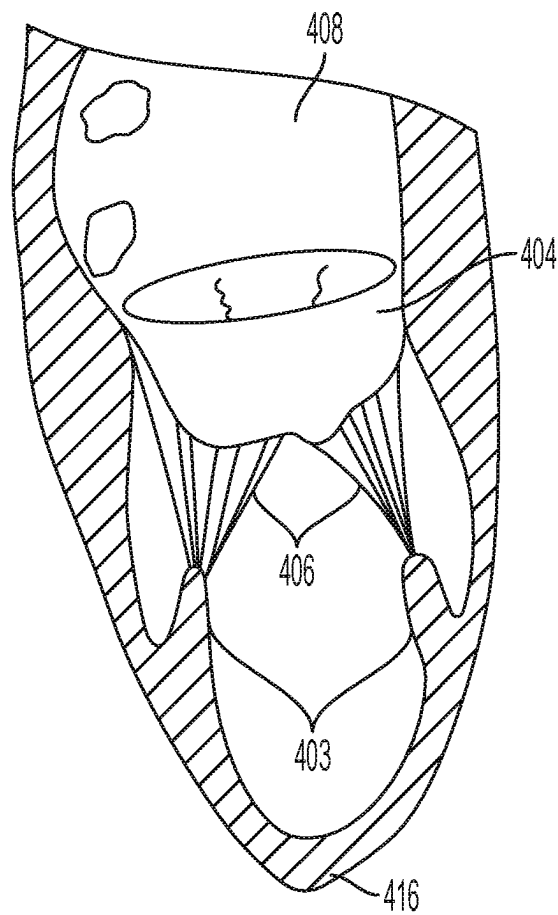


FIG. 4

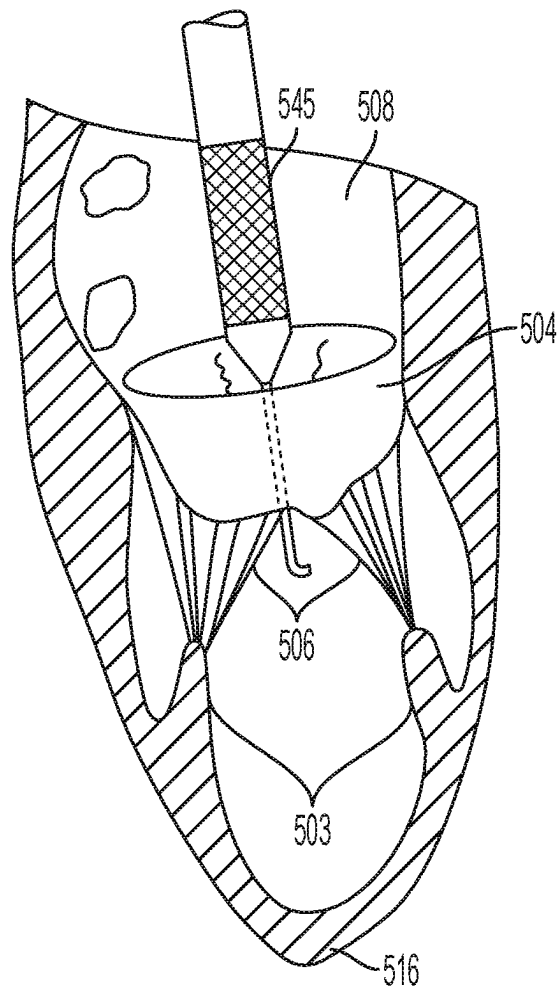


FIG. 5

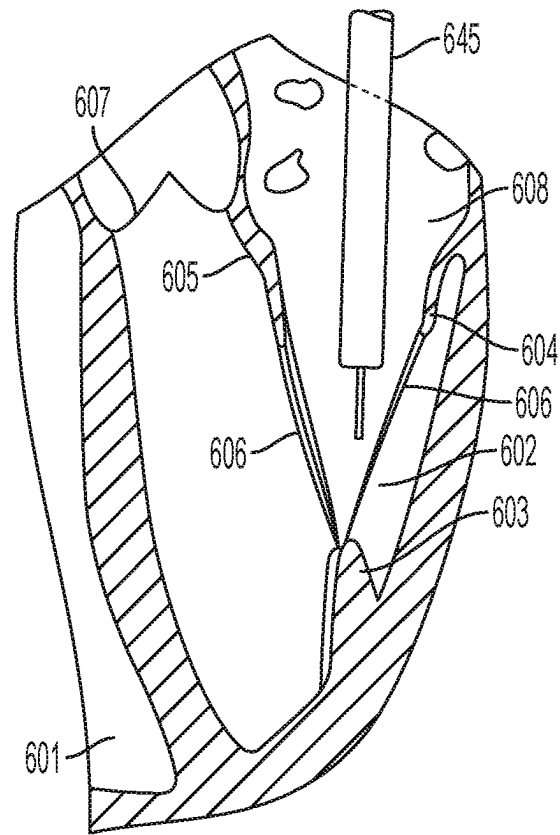


FIG. 6

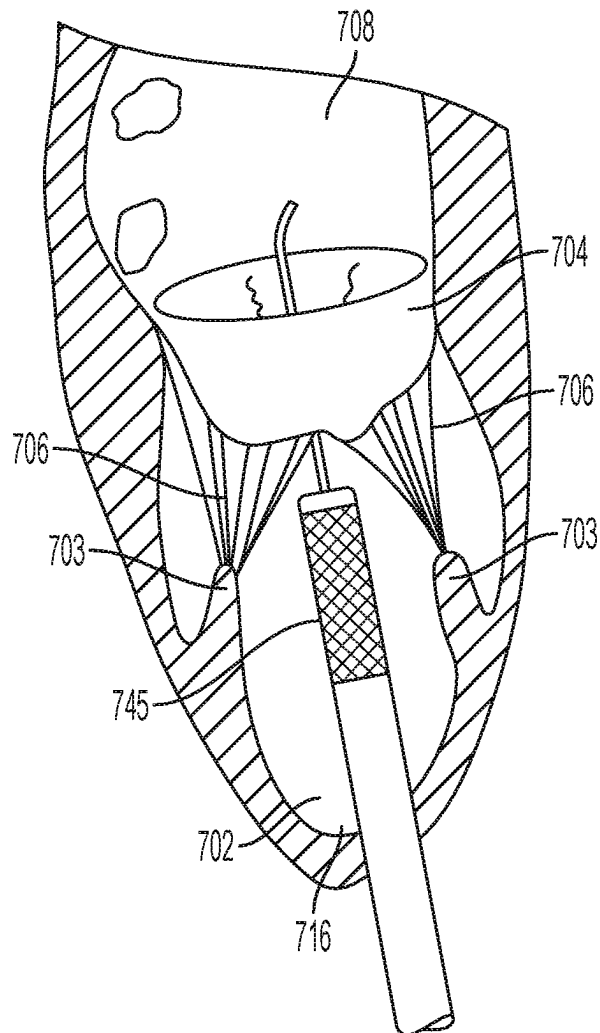


FIG. 7

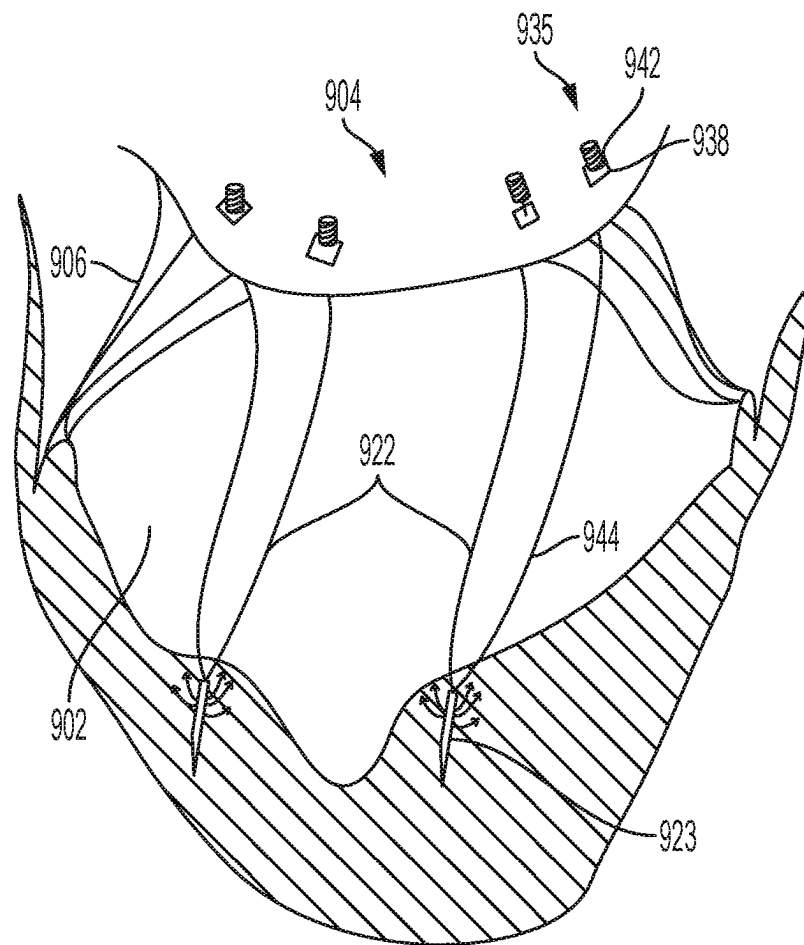


FIG. 9

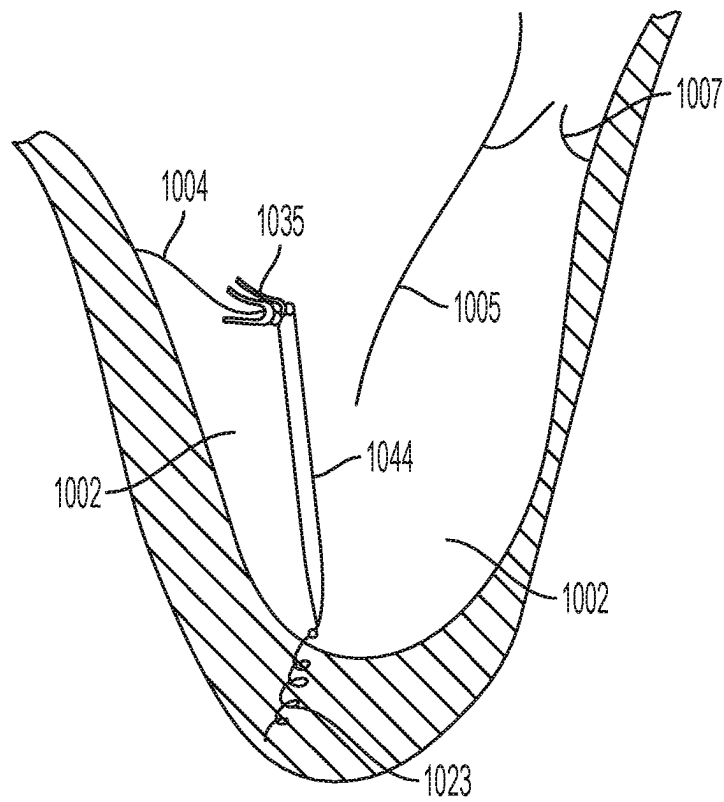


FIG. 10

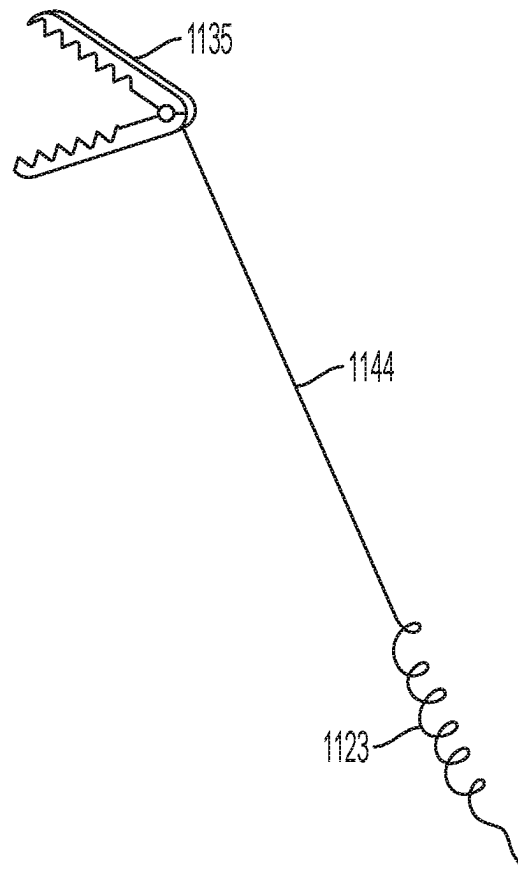


FIG. 11

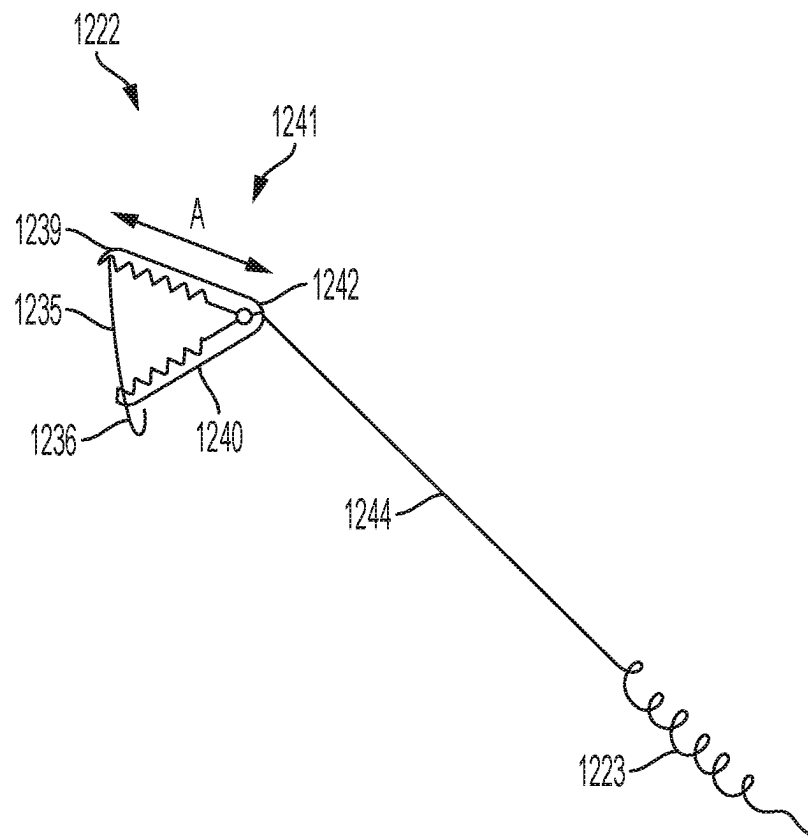


FIG. 12

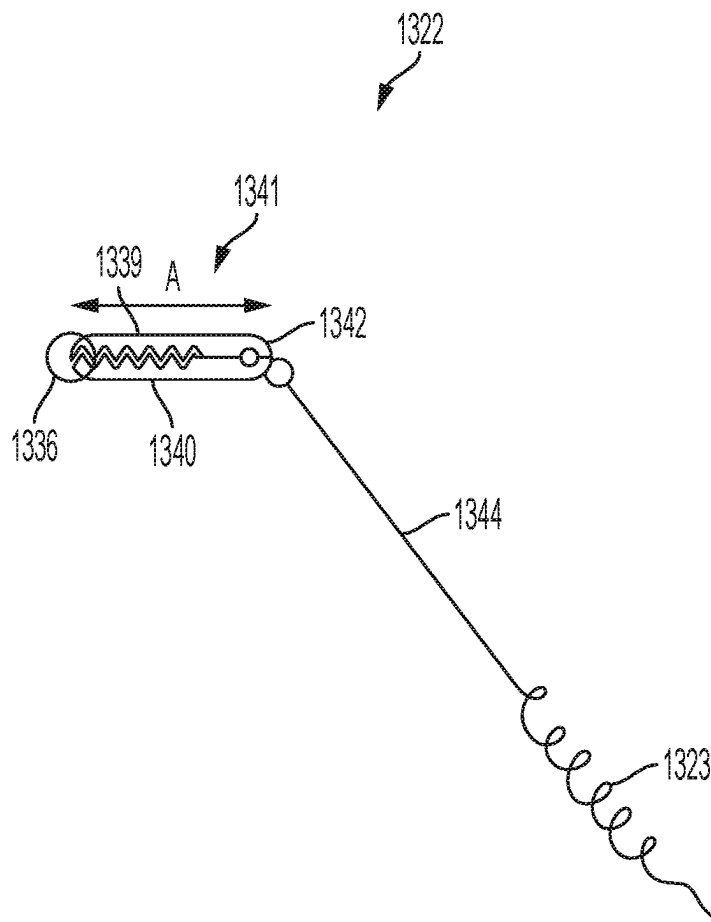


FIG. 13

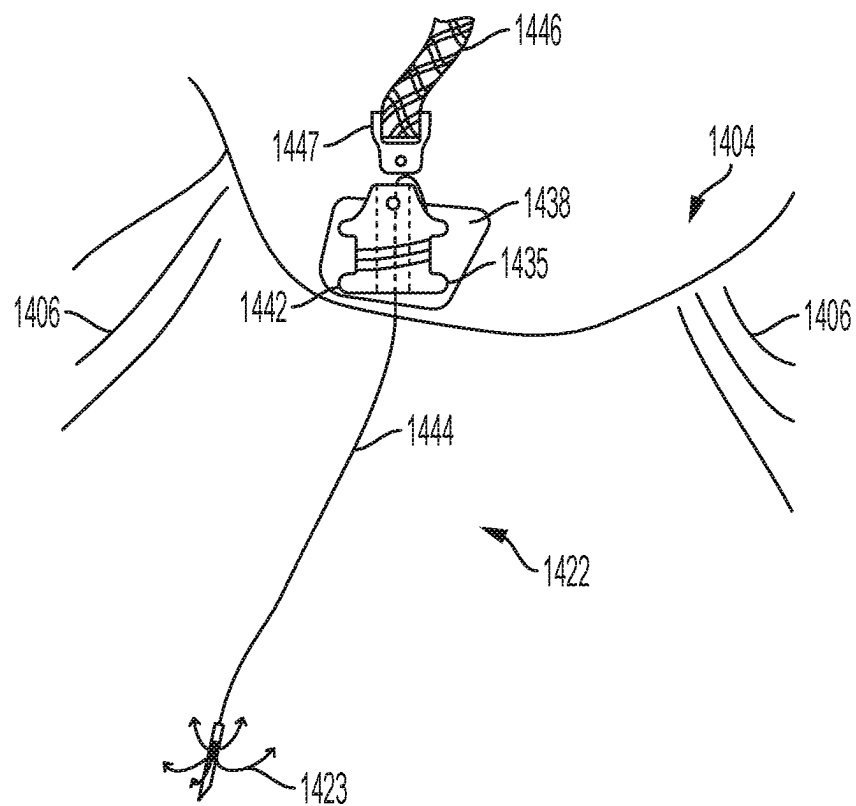


FIG. 14

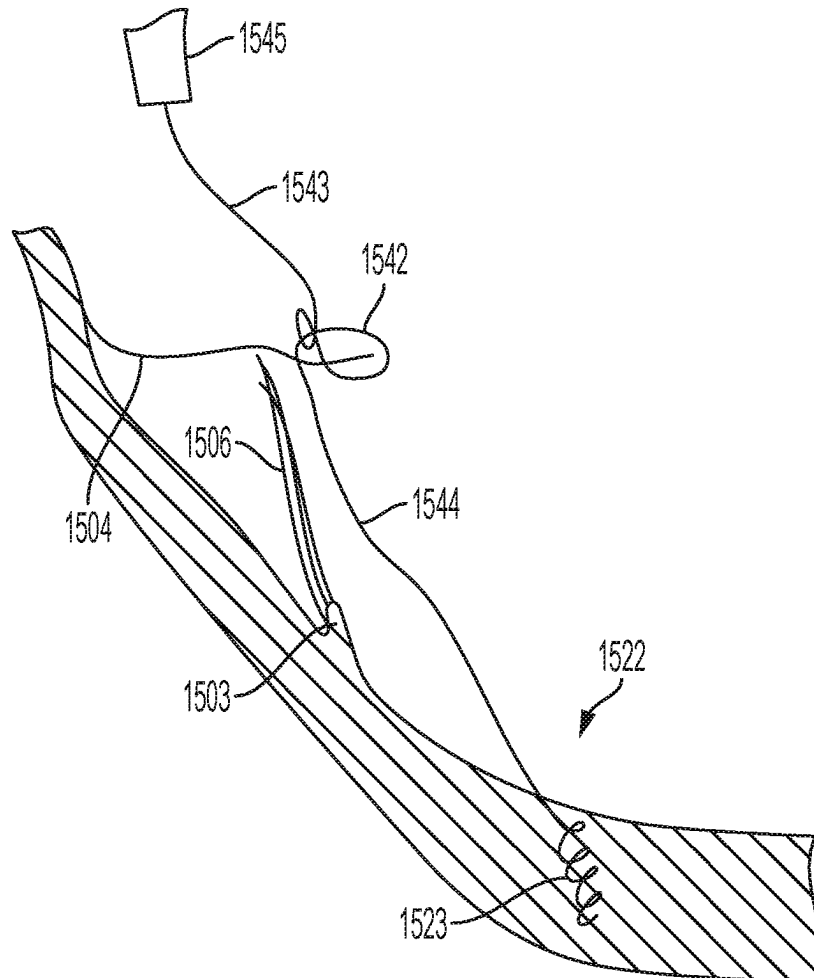


FIG. 15

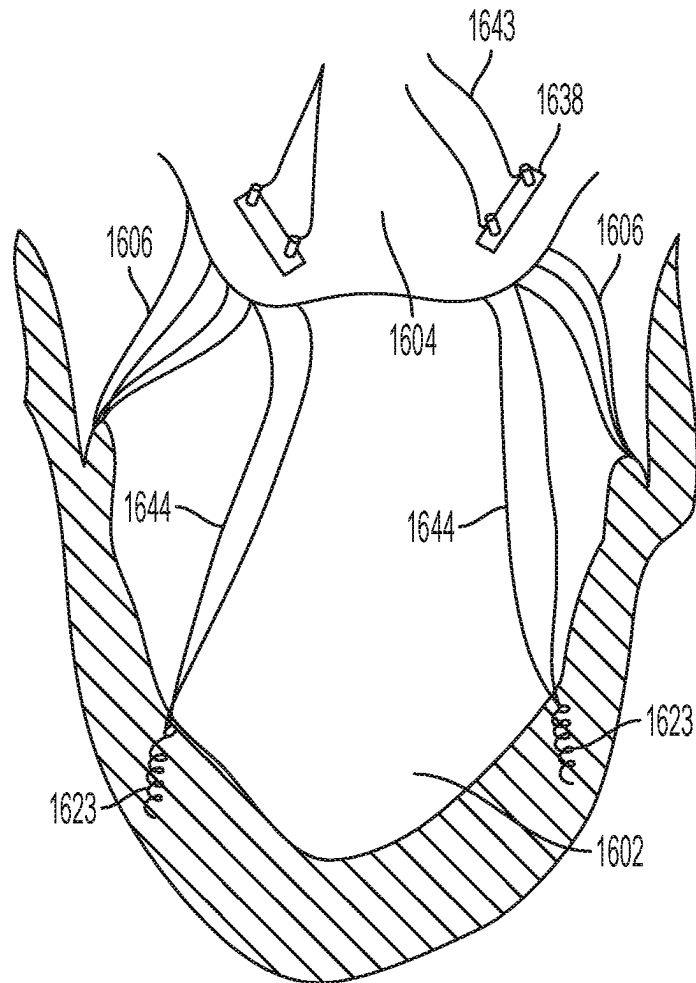


FIG. 16

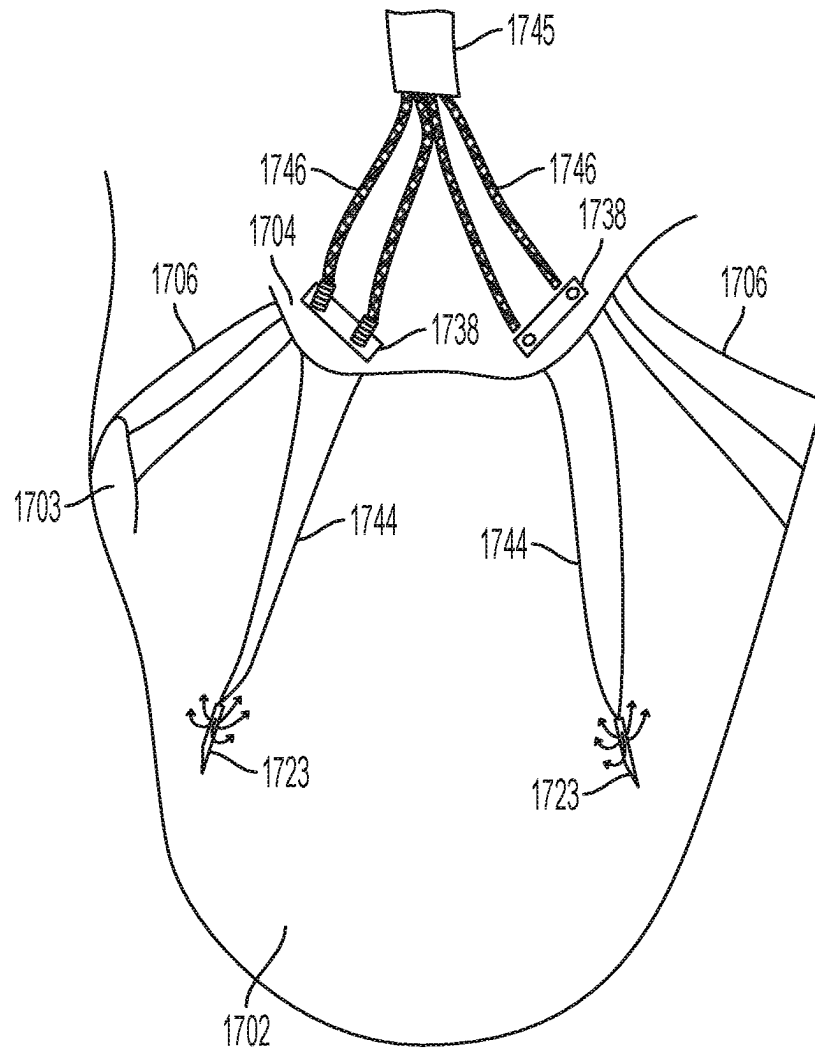


FIG. 17

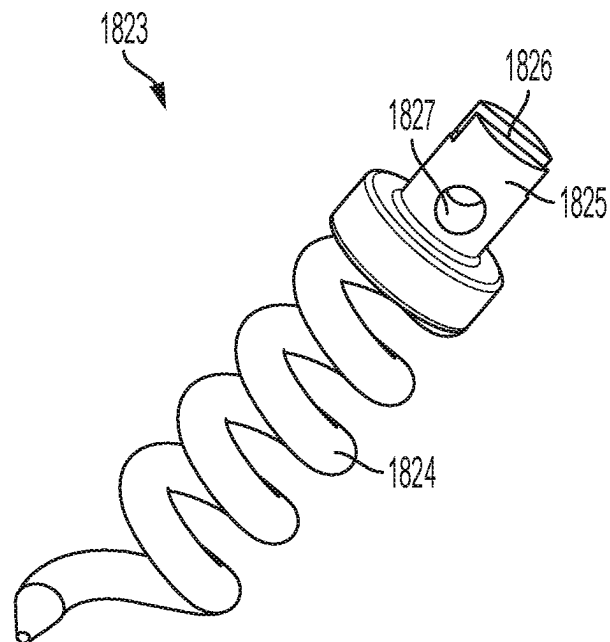


FIG. 18

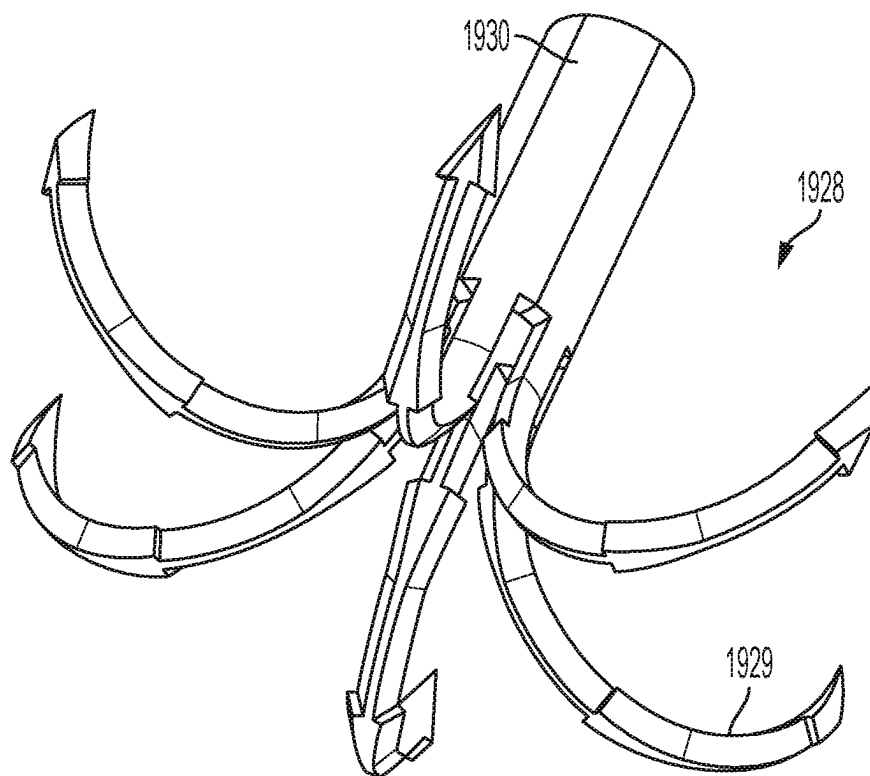


FIG. 19

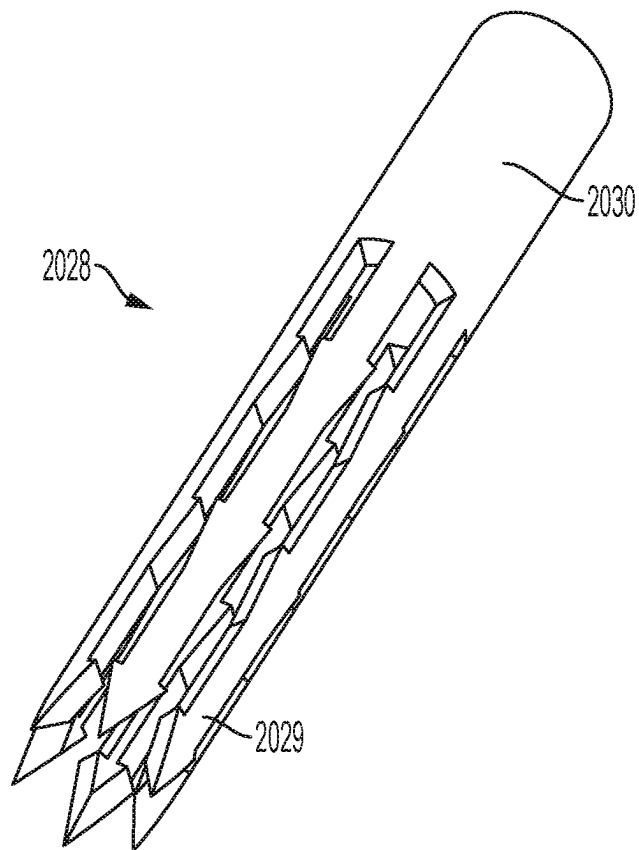


FIG. 20

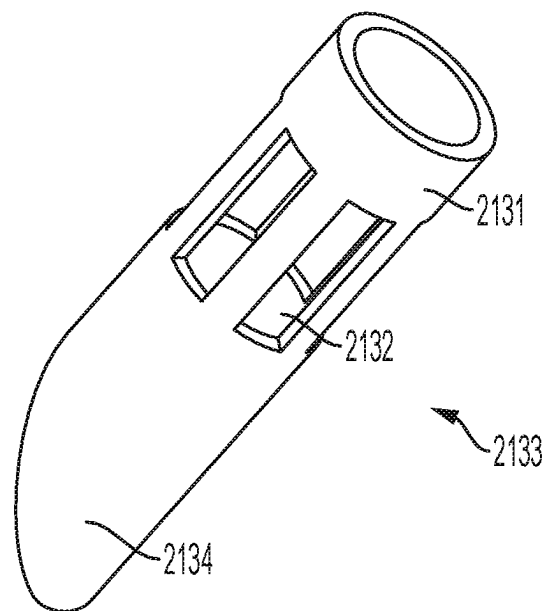


FIG. 21

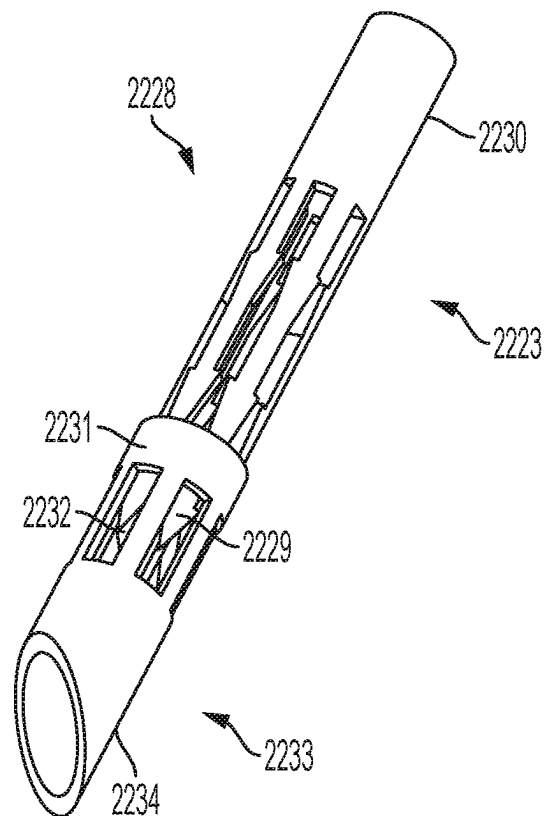


FIG. 22

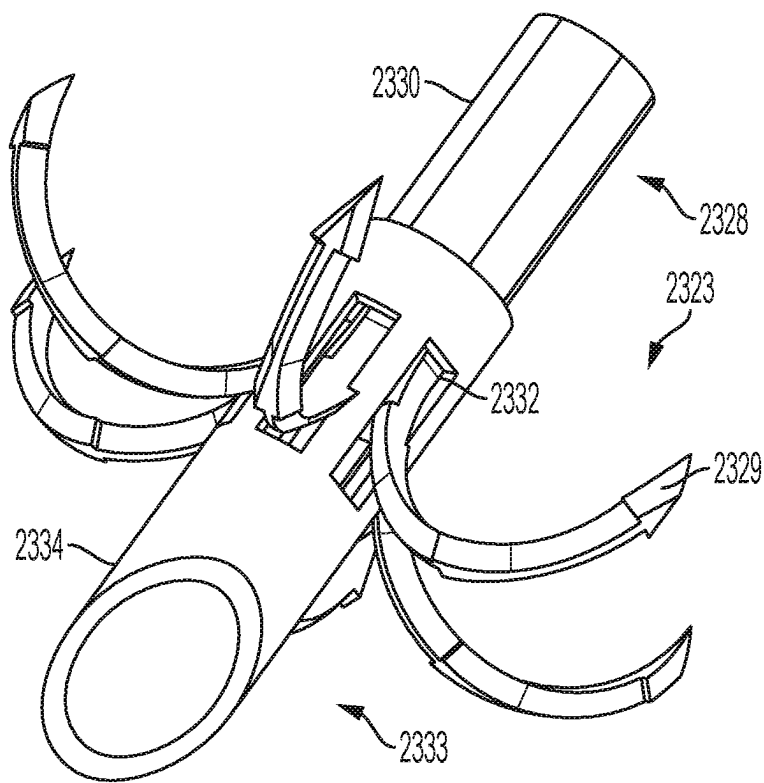


FIG. 23

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SYSTEMS AND METHODS FOR DELIVERY OF CHORDAE REPLACEMENT SYSTEM

PRIORITY

The present application is a national stage filing under 35 U.S.C. § 371 of International Patent Application No. PCT/US2020/037296, filed Jun. 11, 2020, which claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 62/859,968, filed on Jun. 11, 2019, titled “SYSTEMS AND METHODS FOR DELIVERY OF CHORDEA REPLACEMENT SYSTEM,” each of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

The present invention relates to implantable devices. The invention is particularly useful in devices implantable by catheter for the treatment of mitral or tricuspid regurgitation. The cause of the regurgitation can be functional, degenerative, or any other reason. The invention could be used for valvular lesions as well.

Mitral regurgitation is a valvular dysfunction that causes blood volume to flow during systole (i.e., left ventricular contraction) from the left ventricle to the left atrium, in contrast to a healthy heart where this direction of flow is blocked by the mitral valve. The reversed flow during systole causes a rise in pressure in the left atrium. Maintaining a normal cardiac output results in increased left ventricle pressure.

Treating patients with mitral or tricuspid regurgitation could require valve repair or replacement in order to reduce or eliminate the regurgitation. For many years, the commonly accepted treatment was surgical repair or replacement of the native valve during open heart surgery. Valve repair is a procedure that may require complementary treatments such as utilizing annuloplasty rings with or without leaflet or chordae repair. In recent years, a trans vascular technique has been developed for introducing these devices in general and implanting an artificial chorda through a catheter in a manner that is less invasive than open heart surgery.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects, features, benefits and advantages of the embodiments described herein will be apparent with regard to the following description, appended claims, and accompanying drawings where:

FIG. 1 depicts a short axis view of the heart in accordance with an embodiment.

FIG. 2 depicts mitral valve leaflets of the heart in accordance with an embodiment.

FIG. 3 depicts a long axis anterior to posterior view of the heart in accordance with an embodiment.

FIG. 4 depicts a long axis commissure to commissure view of the heart in accordance with an embodiment.

FIG. 5 depicts an illustrative transatrial approach in a commissure to commissure view of the heart in accordance with an embodiment.

FIG. 6 depicts an illustrative transatrial approach in an anterior to posterior view of the heart in accordance with an embodiment.

FIG. 7 depicts an illustrative transapical approach in a commissure to commissure view of the heart in accordance with an embodiment.

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FIG. 8 depicts an illustrative transfemoral approach in an anterior to posterior view of the heart in accordance with an embodiment.

FIG. 9 depicts an illustrative posterior leaflet with a plurality of artificial chordae being implanted in accordance with an embodiment.

FIG. 10 depicts a side view of a posterior leaflet with an artificial chordae being implanted in accordance with an embodiment.

FIG. 11 depicts an illustrative artificial chordae with a leaflet capture mechanism in accordance with an embodiment.

FIG. 12 depicts an illustrative open leaflet capture configuration on an artificial chordae in accordance with an embodiment.

FIG. 13 depicts an illustrative closed leaflet capture configuration on an artificial chordae in accordance with an embodiment.

FIG. 14 depicts an illustrative pulley-based attachment and tuning mechanism in accordance with an embodiment.

FIG. 15 depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment.

FIG. 16 depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment.

FIG. 17 depicts another illustrative anchor-based attachment and tuning mechanism in accordance with an embodiment.

FIG. 18 depicts an illustrative anchor subassembly for an artificial chordae in accordance with an embodiment.

FIG. 19 depicts an illustrative inner member of an anchor subassembly in a deployed configuration in accordance with an embodiment.

FIG. 20 depicts an illustrative inner member of an anchor subassembly in a non-deployed configuration in accordance with an embodiment.

FIG. 21 depicts an illustrative outer member of an anchor subassembly in accordance with an embodiment.

FIG. 22 depicts an illustrative anchor subassembly in a non-deployed configuration in accordance with an embodiment.

FIG. 23 depicts an illustrative anchor subassembly in a deployed configuration in accordance with an embodiment.

DETAILED DESCRIPTION

This disclosure is not limited to the particular systems, devices, and methods described, as these may vary. The terminology used in the description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.

As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. Nothing in this disclosure is to be construed as an admission that the embodiments described in this disclosure are not entitled to antedate such disclosure by virtue of prior invention. As used in this document, the term “comprising” means “including, but not limited to.”

When the term “valvular apparatus” is used in different variations, it may refer to the mitral valve apparatus and/or to the tricuspid valve apparatus and include the leaflets, the chordae, and/or the papillary muscles.

In an embodiment, an artificial chordae device is delivered to the target site, the mitral or tricuspid valve apparatus, through a catheter while the artificial chordae and their

anchoring accessories are housed within a low diameter shaft which is correctly positioned and anchored to the functional configuration geometry and location. The catheter may be advanced to the target site through the vascular system. In an embodiment, the catheter may be advanced from the femoral vein or artery. In alternate embodiments, the catheter may be advanced from any blood vessel that allows access to the target site. In some embodiments the catheter may be advanced transapically, where a catheter is advanced through a small incision made in the chest wall and then through the apex. In some embodiments, the catheter may be advanced transatrially, where a catheter is advanced through a small incision made in the chest wall and then through the left or right atrium.

The artificial chordae device may include one or more of a variety of attachment methods for attachment to the native leaflet and myocardium. The artificial chordae device may include one or more of a variety of locking mechanisms between the leaflet capture subassembly and the native leaflet. In some embodiments, the artificial chordae device may include barbs to prevent rocking and device movement in relation to the leaflet upon insertion. The artificial chordae device may include one or more of a variety of tuning mechanisms. In addition, the artificial chordae device may include one or more of a variety of features that lock the tuning mechanism. Examples of the above-identified variations will be described further herein, although the described variations are illustrative only and are not meant to be limiting.

The illustrated example embodiments will be best understood by reference to the figures. The following description is intended only by way of example and illustrates certain example embodiments.

FIG. 1 depicts a short axis view of the heart displaying the four valves: the mitral valve **121**, the tricuspid valve **109**, the pulmonary valve **110**, and the aortic valve **107**. As shown, the mitral valve **121** includes the posterior leaflet **104** and the anterior leaflet **105**, which will be discussed further herein in relation to various embodiments.

FIG. 2 depicts a perspective of the mitral valve commonly known as the “surgical view” which identifies individual segments of the native leaflets of the mitral valve **221**. As depicted in FIG. 2, individual segments (A1, A2, A3) of the anterior leaflet **205** and individual segments (P1, P2 and P3) of the posterior leaflet **204** are identified. The lateral commissure **219** and medial commissure **220** are also noted for reference.

FIG. 3 depicts a left ventricular outflow tract view, anterior to posterior, of the heart which shows how the anterior **305** and posterior **304** leaflets are attached to the papillary muscles **303** by the chordae **306**. FIG. 3 further depicts the left ventricle **302** and the right ventricle **301** separated by the intraventricular septum **312**, the apex **316**, the left atrium **308**, and the aortic valve **307**. Within the left ventricle **302**, the posterior wall **311**, mitral groove **314**, and aortic-mitral continuity **315** are specifically identified.

FIG. 4 depicts a perspective of the left side of the heart in commissure to commissure view showing the left atrium **408**. Similar to FIG. 3, FIG. 4 depicts the attachment of the posterior leaflet **404** to the myocardium **403** via the chordae **406** near the apex of the heart **416**.

FIG. 5 depicts an illustrative transatrial approach of a chordae repair device to the mitral valve apparatus in a commissure to commissure view of the heart in accordance with an embodiment. When using the transatrial approach, a catheter **545** is introduced into the left atrium **508** directly. In addition, the transatrial approach may pass by or cross-

over the mitral valve and allow implantation of a chordae repair device between the posterior leaflet **504** and/or the papillary muscles **503** to repair the chordae **506**.

FIG. 6 depicts an illustrative transatrial approach of a chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment. The catheter **645** is introduced into the left atrium **608** directly at the native valve comprising the anterior **605** and posterior **604** leaflets. Using this approach, implantation of aa chordae repair device may be made to the leaflets **604**, **605** and the papillary muscles **603** in repair of the chordae **606** within the left ventricle **602**. FIG. 6 further depicts the right ventricle **601** and aortic valve **607** for reference.

FIG. 7 depicts an illustrative transapical approach of a chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment. Using the transapical approach, the catheter **745** is introduced between the left atrium **708** and the ventricle **702**. The transapical approach is a direct approach where the catheter **745** is introduced into the left ventricle across the apex **716** to allow implantation of the chordae repair device to the valvular apparatus to anchor between the leaflet **704** (e.g., posterior leaflet, interior leaflet, etc.) and the papillary muscles **703** to repair the chordae **706**.

FIG. 8 depicts an illustrative transfemoral approach of the chordae repair device to the mitral valve apparatus in an anterior to posterior view of the heart in accordance with an embodiment. The transfemoral approach is a percutaneous approach where the catheter **845** is introduced into the femoral vein and advanced along the inferior vena cava to the right atrium, across the septum **812**, to allow implantation of a chordae repair device to the valvular apparatus including the anterior **805** and posterior **804** leaflets, papillary muscles **803**, and chordae **806**. The aortic valve **807** is also shown for reference.

FIG. 9 depicts an illustrative posterior leaflet with a plurality of artificial chordae being implanted in accordance with an embodiment. As shown in FIG. 9, an artificial chordae device **922** is implanted at a plurality of locations on the posterior leaflet **904** of the left ventricle **902**. The artificial chordae device **922** may be used to repair chordae **906** that failed due to, for example and without limitation, leaflet prolapse, flail, or other pathology. In some embodiments, the artificial chordae device **922** comprises a plurality of artificial chordae **944**, an anchor subassembly **923** configured to attach to the myocardium, and a leaflet capture subassembly **935**, with a tuning mechanism **942** and stress relief pad **938**, configured to attach to the native leaflet. The implantation of one or more artificial chordae devices **922** may be performed serially or in parallel. Each artificial chordae device **922** may be tuned separately using the corresponding tuning mechanism **942**. The one or more artificial chordae devices **922** may be visualized using, for example and without limitation, endoscopy, fluoroscopy, a CT scan, an MRI scan, and/or vital signals to assist in optimizing the result.

FIG. 10 depicts a side view of a posterior leaflet with an artificial chordae device being implanted in accordance with an embodiment. As shown in FIG. 10, one or more leaflet capture assemblies **1035** may be attached to the posterior leaflet **1004** in the left ventricle **1002** from a side view. The one or more leaflet capture assemblies **1035** may include a mechanism to ensure connectivity and proper tension. In some embodiments, the connectivity and tension may be tunable (e.g., separately or in combination) during and/or after deployment and attachment. An anchor assembly **1023**

may be used to attach the artificial chordae **1044** to the myocardium. The anterior leaflet **1005** and aortic valve **1007** are shown for reference.

FIG. **11** depicts an illustrative artificial chordae with a leaflet capture mechanism in accordance with an embodiment. An artificial chordae device comprises an artificial chordae **1144**, a leaflet capture subassembly **1135**, and an anchor subassembly **1123**.

FIGS. **12** and **13** depict the leaflet capture subassembly **1235** of the artificial chordae, which is composed of an upper jaw **1239**, a lower jaw **1240**, and a lock feature **1236** to assure proper attachment to the native leaflet. The jaws **1239**, **1240** capture the leaflet and have a depth of 2-8 mm, shown as dimension A **1241**. In an embodiment, the geometry and the attachment depth of the jaws **1239**, **1240** may be designed to prevent damage and unintentional detachment from the native leaflet. The artificial chordae **1244** may be attached to the jaws **1239**, **1240** at the tuning mechanism **1242**. The artificial chordae **1244** may be attached to a muscle using an anchor subassembly **1223**. FIG. **12** depicts the artificial chordae **1244** with the leaflet capture subassembly **1235** in an open configuration.

FIG. **13** depicts the artificial chordae **1344** with the leaflet capture subassembly **1335** in a closed configuration. As shown in FIG. **13**, the locking mechanism **1336** is engaged. As shown, dimension A **1341** represents the attachment depth of the leaflet capture subassembly **1335** to the native leaflet. In some embodiments, the tension of the artificial chordae **1344** may be adjusted at the tuning mechanism **1342**. In some embodiments, the artificial chordae **1344** may be attached to a muscle using an anchor subassembly **1323**.

FIG. **14** depicts an illustrative pulley-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **14**, the artificial chordae device **1422** may include a pulley tuning mechanism. In some embodiments, the tuning mechanism may include a pulley **1442** that may be controlled by a torque cable **1446** resident in a delivery catheter. In some embodiments, the torque cable **1446** may be attached to the pulley **1442** via a connection feature **1447**. In some embodiments, the leaflet capture assembly **1435** may include a stress relief pad **1438** that enables forces resulting from attachment to the native leaflet **1404** to be distributed. In such embodiments, distributing the attachment forces may prevent damage to the native leaflet **1404** and encapsulate the assembly **1435** after healing. In some embodiments, the artificial chordae **1444** may be attached to the muscle using a barbed anchor subassembly **1423**. The artificial chordae **1444** may be tuned so that the artificial chordae replace the operation of the damaged chordae **1406**.

FIG. **15** depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **15**, an artificial chordae device **1522** may include a tuning mechanism that includes a knot **1542**. In some embodiments, the knot **1542** may be adjustable and may be used to tune the artificial chordae device **1522** to the correct length. In some embodiments, the knot **1542** may be tuned to the correct length through the delivery catheter **1545**. In some embodiments, the knot **1542** may be locked to a final length through movement of a proximal section of a tightening suture **1543**. In some embodiments, the artificial chordae device **1522** may be attached to the posterior leaflet **1504** and anchored via a screw-based anchor **1523** to the papillary muscles **1503** to repair the damaged chordae **1506**.

FIG. **16** depicts an illustrative knot-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **16**, a plurality of artificial chordae **1644** are implanted in the left ventricle **1602** to properly repair the

damaged chordae **1606**. In some embodiments, each artificial chordae may be attached to the posterior leaflet **1604** via a suture **1643** and anchored to the myocardium via a screw-based anchor **1623**. In some embodiments, the length of each of the artificial chordae **1644** may be adjusted independently by adjusting, for example, the corresponding knot-based tuning mechanism. In some embodiments, adjustments to the tuning mechanism may be performed from the delivery catheter.

In some embodiments, multiple devices may be attached to the posterior leaflet or the anterior leaflet. In such embodiments, an operator may control the distance between adjacent attachments to leaflet mechanisms. In another embodiment, one or more pads may be used for stress relief of the attachment. In some embodiments, the pads may be large patches that could support a plurality of attachment mechanisms **1638**. In alternate embodiments, smaller patches may be used that only support a single mechanism **1638**.

FIG. **17** depicts another illustrative anchor-based attachment and tuning mechanism in accordance with an embodiment. As shown in FIG. **17**, artificial chordae **1744** may be implanted to replace damaged chordae **1706** in the left ventricle **1702**. In some embodiments, the artificial chordae **1744** may be attached to the posterior leaflet **1704** in pairs on stress relief pads **1738**. In some embodiments, the artificial chordae **1744** may be anchored to the myocardium below the papillary muscles **1703**. In some embodiments, the tuning mechanism may include one or more torque cables **1746** controlled via the delivery catheter **1745**. In some embodiments, a barbed assembly **1723** may be used to anchor the artificial chordae **1744**.

The length of each of the artificial chordae may be from about 20 mm to about 120 mm. For example, the length of an artificial chordae may be tuned to 20 mm, 30 mm, 40 mm, 50 mm, 60 mm, 70 mm, 80 mm, 90 mm, 100 mm, 110 mm, 120 mm, or to a length within a range between any two of these endpoints. The variation in the length of the artificial chordae may be used to address variations in ventricle size in a patient population as well as to allow the operator to make a decision as to where to anchor the assembly. For example, different operators may choose to anchor the assembly near to the apex, while other operators may choose to anchor the assembly near to the papillary muscle. Such decisions may be based on the anatomy of the patient and/or operator preference.

In some embodiments, the design of a chordae replacement device may allow loading the device into a low profile shaft having an outer diameter that is, for example, less than or equal to 13 mm. In some embodiments, the chordae replacement device and the associated delivery system may enable delivery of a plurality of devices, such as the leaflet capture assembly, anchor assembly, and artificial chordae. In some embodiments, the plurality of devices may be delivered consecutively or simultaneously. In some embodiments, the plurality of devices may be housed in parallel inside the catheter prior to delivery. In some embodiments, the plurality of devices may be housed consecutively within the catheter prior to delivery. In some embodiments, only one of the plurality of devices may be loaded into the catheter at a time. In some embodiments, the control mechanism (e.g., the torque cable **1746**) may be retained within the catheter.

The leaflet capture assembly may be configured to attach to the native leaflets of the tricuspid valve or the mitral valve. When used in the mitral position, the leaflet capture assembly may lean in against the edges of the mitral leaflets, anterior or posterior, according to the area that requires

treatment. In some embodiments, the attachment mechanism may enable tuning of the tension on the artificial chordae that attaches leaflets to the myocardium. In some embodiments, the attachment mechanism may enable precise positioning with respect to the leaflet. In some embodiments, the attachment mechanism may include a padded surface to distribute the forces and/or to encourage encapsulation of the assembly into the leaflet.

One of ordinary skill in the art will be aware that any combination of the various components described herein and equivalents may be used for the construction of a particular device based on this disclosure. In other words, the devices depicted herein are merely illustrative of the types of devices that may be constructed according to the teachings of this disclosure and are not meant to be limited to these illustrative embodiments.

FIG. 18 depicts an illustrative anchor subassembly for an artificial chordae in accordance with an embodiment. The anchor assembly 1823 may be configured to attach to the myocardium. In some embodiments, the anchor assembly 1823 may include a self-expanding stent comprising a shape memory material (e.g., Nitinol). In some embodiments, the anchor assembly 1823 may be cut from a tube or sheet and/or use a pattern that allows crimping and expanding. In one embodiment, the anchoring assembly 1823 may be constructed from a wire formed into a coiled geometry configured to be screwed into the tissue. The coiled portion 1824 of the anchor assembly 1823 may be configured to penetrate into tissue and maintain an anchoring force during implantation and in vivo life expectancy.

In some embodiments, the anchor assembly 1823 may feature an interface portion 1825 that is configured to enable an attachment with a catheter during implantation. The interface portion 1825 may include an attachment 1826 to the delivery catheter that allows transmission of torque forces. The torque forces may be applied to the attachment 1826 to rotate the anchor assembly 1823 and enable the coiled portion 1824 to penetrate into tissue. In some embodiments, axial forces may also be applied to advance the anchor assembly 1823 into or retract the anchor assembly from tissue. In some embodiments, the interface portion 1825 may include one or more holes 1827 configured to attach artificial chordae to the anchor assembly 1823. Illustrative coil-based anchor assemblies are further depicted in FIGS. 10-13, 15, and 16.

FIG. 19 depicts an illustrative inner member of an anchor subassembly in a deployed configuration in accordance with an embodiment. In some embodiments, an anchor subassembly may include an outer member (depicted in FIG. 21) and an inner member 1928. The inner member may include a tube having a specific pattern that allows two different configurations as depicted in FIGS. 19 and 20. As shown in FIG. 19, a plurality of anchors 1929 may be positioned on a first end of the anchor subassembly 1928. The plurality of anchors 1929 may expand radially to anchor into tissue. In some embodiments, the opposite end 1930 of the anchor subassembly 1928 may provide an interface for attachment to the artificial chordae.

FIG. 20 depicts the illustrative inner member of the anchor subassembly of FIG. 19 in a non-deployed configuration. As shown in FIG. 20, the anchors 2029 of the anchor subassembly 2028 may be aligned parallel with the interface 2030 of the anchor subassembly. The anchor 2029 may be in the non-deployed configuration when housed within the outer member of the anchor subassembly as discussed further in reference to FIGS. 21 and 22.

In some embodiments, the inner member of the anchor subassembly may be constructed from one or more metals and/or an alloy. In one embodiment, the inner member is constructed from one or more metals that are biocompatible and are configured to permit the device to transform between the two configurations. For example, the inner member may comprise one or more of cobalt, chrome, stainless steel, or Nitinol.

FIG. 21 depicts an illustrative outer member of an anchor subassembly in accordance with an embodiment. As shown in FIG. 21, an outer member 2133 of an anchor assembly may comprise a tube with a plurality of windows 2132. In some embodiments, each of the plurality of windows may be located radially. Different window locations are contemplated within the scope of this disclosure. In some embodiments, the outer member 2133 may include a distal portion 2134 having a sharp edge. The sharp edge of the distal portion 2134 may be configured to enable the outer member to penetrate into tissue. In some embodiments, the outer member 2133 may further include a proximal section 2131. In some embodiments, the proximal section 2131 of the outer member 2133 may be used for attachment to the delivery system and/or to the artificial chordae.

In some embodiments, the outer member 2133 of the anchor subassembly may be constructed from one or more metals and/or an alloy. In one embodiment, the outer member 2133 is constructed from one or more metals that are biocompatible. For example, the outer member 2133 may comprise one or more of cobalt, chrome, stainless steel, or Nitinol. In some embodiments, the outer member 2133 may comprise a plastic, such as polyetheretherketone, other materials, or other combinations of materials that are biocompatible.

FIGS. 22 and 23 depict an illustrative anchor subassembly in a non-deployed configuration 2223 and a deployed configuration 2323, respectively, in accordance with an embodiment. In some embodiments, the inner member 2228/2328 may include a plurality of anchors 2229/2329 and the artificial chordae interface 2230/2330. In some embodiments, the inner member may be inserted into the outer member 2233/2333. In some embodiments, the outer member 2233/2333 may include a plurality of anchor windows 2232/2332, a control interface 2231 to the delivery system, and a distal end 2234/2334.

In some embodiments, transitioning between the non-deployed configuration and the deployed configuration may be controlled via the delivery catheter by advancing, retracting, and/or rotating the catheter. In some embodiments, transitioning between the two configurations may further be performed with the outer member. For example, linear movement of the inner member 2228/2328 with respect to the outer member 2233/2333 may cause the plurality of anchors 2229/2329 to transition between the two configurations. In some embodiments, movement of the inner member 2228/2328 may cause the outer member 2233/2333 to deflect the anchors through the windows 2232/2332 mechanically. In some embodiments, the inner member 2228/2328 may be manufactured from a shape memory material, such as Nitinol, and movement of the inner member with respect to the outer member 2233/2333 may cause the anchors to transition to the deployed configuration. In some embodiments, the control interface 2231 may be designed to allow an easy and safe release after the device is anchored to the tissue.

In some embodiments, the inner member may be manufactured by machining, grinding, and/or laser cutting. In some embodiments, the distal ends 2234/2334 of the

anchors **2229/2329** may be sharpened or grinded to enable tissue penetration. In some embodiments, the anchors **2229/2329** may be aligned in a specific direction. In some embodiments, the anchors **2229/2329** may be aligned in two opposing directions. In some embodiments, the anchors **2229/2329** may be aligned in a plurality of directions.

Any number of anchors **2229/2329** may be used within the scope of this disclosure. In some embodiments, the distal ends **2234** may be inside the corresponding windows **2232** to assure deployment through the windows when the inner member **2228** is within the outer member **2233** in the non-deployed configuration. In some embodiments, the inner member **2228** may be stored within the catheter and advanced into the outer member **2233** separately. In some embodiments, the inner diameter of the outer member **2233** may be in a range of about 0.8 mm to about 4 mm. In some embodiments, the outer diameter of the inner member **2228** may be in a range of about 0.8 mm to about 4 mm, but in any event is less than the inner diameter of the outer member **2233** to allow movement of the inner member therein.

For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (for example, “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations.

In addition, even if a specific number of an introduced claim recitation is explicitly recited, those skilled in the art will recognize that such recitation should be interpreted to mean at least the recited number (for example, the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, et cetera” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, and C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). In those instances where a convention analogous to “at least one of A, B, or C, et cetera” is used, in general such a construction is intended in the sense one having skill in the art would understand the convention (for example, “a system having at least one of A, B, or C” would include but not be limited to systems that have A alone, B alone, C alone, A and B together, A and C together, B and C together, and/or A, B, and C together, et cetera). It will be further understood by those within the art that virtually any disjunctive word and/or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” will be understood to include the possibilities of “A” or “B” or “A and B.”

As will be understood by one skilled in the art, for any and all purposes, such as in terms of providing a written description, all ranges disclosed herein also encompass any and all possible subranges and combinations of subranges thereof.

Any listed range can be easily recognized as sufficiently describing and enabling the same range being broken down into at least equal halves, thirds, quarters, fifths, tenths, et cetera. As a non-limiting example, each range discussed herein can be readily broken down into a lower third, middle third and upper third, et cetera. As will also be understood by one skilled in the art all language such as “up to,” “at least,” and the like include the number recited and refer to ranges that can be subsequently broken down into subranges as discussed above. Finally, as will be understood by one skilled in the art, a range includes each individual member. Thus, for example, a group having 1-3 cells refers to groups having 1, 2, or 3 cells. Similarly, a group having 1-5 cells refers to groups having 1, 2, 3, 4, or 5 cells, and so forth.

Various of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations or improvements therein may be subsequently made by those skilled in the art, each of which is also intended to be encompassed by the disclosed embodiments.

What is claimed is:

1. A method of repairing a native chordae of a patient using an artificial chordae, the method comprising: inserting the artificial chordae into the patient using a delivery system, wherein the artificial chordae is inserted in a non-deployed configuration; delivering the artificial chordae to a desired position within the patient; causing the artificial chordae to transition from the non-deployed configuration to a deployed configuration; anchoring the artificial chordae to a myocardium of the patient via an anchoring assembly comprising an inner member comprising a barbed anchor and an outer member comprising a plurality of windows, the inner member positioned within the outer member; attaching the artificial chordae to a leaflet of the native chordae at an attachment location via a leaflet capture subassembly coupled to the artificial chordae, the leaflet capture subassembly comprising an upper jaw and a lower jaw configured lock the leaflet between the upper jaw and the lower jaw; and tuning the artificial chordae to a desired tension, wherein the artificial chordae is anchored to the myocardium by deploying the barbed anchor of the inner member through the plurality of windows of the outer member and into the myocardium to secure the artificial chordae to the myocardium.
2. The method of claim 1, further comprising delivering one or more additional devices to the desired position.
3. The method of claim 2, wherein the one or more additional devices comprise the anchoring assembly and the leaflet capture subassembly.
4. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered via a single catheter.
5. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered in parallel within a single catheter.
6. The method of claim 2, wherein the artificial chordae and the one or more additional devices are delivered in series within a single catheter.
7. The method of claim 2, further comprising tuning the one or more additional devices and the artificial chordae simultaneously.

8. The method of claim 1, further comprising installing a stress relief pad at the attachment location on the leaflet of the native chordae, wherein the stress relief pad is configured to lower stress on the leaflet and assist in healing thereof.

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9. The method of claim 8, further comprising delivering one or more additional devices to the desired position, wherein the artificial chordae and the one or more additional devices are configured to share use of the stress relief pad.

10. The method of claim 1, wherein the artificial chordae is tuned via a tuning mechanism, the tuning mechanism comprising a pulley interfaced to a torque cable.

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11. The method of claim 1, wherein the artificial chordae is tuned via a tuning mechanism, the tuning mechanism comprising a knot in a tightening suture that is configured to be adjusted to a desired length and locked to the desired length through movement of a proximal section of the tightening suture.

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12. The method of claim 1, wherein the barbed anchor is configured to be delivered to an anchoring location in a non-deployed configuration and then deployed radially into a tissue of the myocardium to secure the artificial chordae thereto.

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13. The method of claim 12, wherein the outer member is configured to:

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lock the barbed anchor in place during delivery; and control deployment of the barbed anchor through movement of the delivery system.

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